

11th CIRP Global Web Conference (CIRPe 2023)

Voice User Interface based control for Industrial machine tools

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To meet productivity goals, machining operations require high levels of efficiency to achieve the desired manufacturing quality in the available time. Additionally, machine tools can lead to safety-critical scenarios due to the high process dynamics and hazardous forces and torques. Therefore, workers need to exhibit special care while interacting with certain kinds of machines. Operational safety can be increased by eradicating the dependence on tangible control of machine tools. Current developments in Artificial Intelligence, especially in natural language processing, lead to increased utilization of speech-based interaction between humans and machines. However, these developments are mainly limited to areas of consumer electronics. Nevertheless, this type of human-machine interaction offers great potential for machining operations. Artificial neural network architectures like Transformer have rapidly emerged as the dominant architecture for natural language processing, overtaking alternative neural architectures such as Convolutional and Recurrent Neural Networks in performance for natural language understanding and natural language generation tasks. The architecture scales with training data and model size, enables efficient parallel training, and captures wide-ranging sequence features. These advancements can be used for enhancing the ease of use and safety of machine tools, which are often safety critical. In this article, a Voice-User-Interface based control of machine tools is conceptually outlined. Such a control can increase the efficiency of the workers, as it eradicates the need of hand usage. It further increases the safety of workers operating the machines and can detect anomalous sounds corresponding to machine failures during the machining process. The voice is first passed through a speech-to-text pipeline for converting the speech signals to relevant text, the text output is mapped to machine commands and passed to the physical machine. The implementation of the setup guarantees the speech-based control – even for safety-critical tasks -of machine tools. Moreover, this paper outlines the benefits and challenges of the developed VUI architecture.

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Peer-review under responsibility of the scientific committee of the 11th CIRP Global Web Conference

Keywords: Voice User Interface; Machine Tools; Artificial Neural Networks; Transformers; Convolutional Neural Networks; Recurrent Neural Networks**1. Motivation**

Machine tools are considered the backbone of the manufacturing industry as their performance deeply affects production efficiency and effectiveness. Industry 4.0 is characterized by advancing digitalization, increasing customer requirements, and individualization. Cyber-Physical Machine Tools (CPMT) are the next generation of Machine Tools that can cater to the needs of Industry 4.0. They provide more autonomy and operational efficiency over their currently prevalent CNC machine counterparts [1]. One of the major components of the conceptualized CPMT is the development of smart human-machine interfaces(HMI). The current CNC machines can facilitate precise and highly automated machining but are limited by their human-machine interface which needs tangible control of a properly trained individual [4]. The HMI is present mainly as a control panel attached to the machine which also limits the

mobility of the operator handling the machine. Advances in Machine Learning especially in the fields of Natural Language Processing (NLP), speech-to-text and text-to-speech have led to the development of smart Voice User Interfaces (VUI) as is evident in some common dialog-based assistants like Amazon Alexa, and Siri from Apple. VUI can prove useful interfaces for interacting with machine tools, as they can mitigate certain major obstacles like non-inclusiveness, non-intuitiveness, and operator immobility posed by conventional HMIs. This article introduces a conceptual architecture of a VUI that can facilitate dialog-based control and interaction with modern-day CNC machine tools.

2. State of the Art

2.1. Natural Language Processing

Natural language processing (NLP) refers to the discipline of interpreting and analyzing natural language in either oral or written format. Any language that humans learn from the environment or use to communicate with each other is defined as natural language [5]. NLP has a rich history starting from Chomsky's classification of natural languages [3] which mainly aimed at rule-based grammars for understanding the syntax of the languages. Due to advances in the field of machine learning algorithms, the current NLP algorithms can easily detect the semantics as well as the intention of spoken as well as written text [3]. NLP is used in a wide range of application areas like Sentiment analysis, Named entity recognition, Part of speech tagging, Machine translation, and Natural language understanding (NLU) [17]. The recently developed Transformer [6] architecture has revolutionized NLP leading to innovative solutions such as GPT [8], BERT [7].

Transformers are simple encoder, and decoder models which consist of multi-head attention, add and normalize, and feed-forward layers. The encoder and decoders can be iteratively added to the transformer architecture. The input data is either passed to the encoder or a decoder dependent on the application. The first implementation of the transformer proved its applicability in the field of NLP which is mainly dominated by sparse data [6]. The network diagram of the first implemented architecture is shown in Fig. 1. Transformer architectures have also been widely used in TTS and STT services. Pre-trained Transformer architectures like BERT can be fine-trained using machine-specific vocabulary for building an NLU module for controlling machine tools in a complex manufacturing environment. Such a VUI can facilitate safe, flexible, and intangible control of machine tools.

2.2. Human-Machine and Voice-User Interfaces

In manufacturing systems human-machine interfaces (HMIs), connector, operator, and machine tool, serve as an orchestration element within a human-machine interaction, and enable bidirectional communication [9]. HMIs consist of different input and output technologies [10]. The input technologies can be classified into tangible and intangible interfaces depending on the physical interaction involved. Intangible input interfaces, e.g., motion capturing or voice recognition, allow for high intuitiveness and easy interaction with machines without requiring a dedicated space in the manufacturing system. However, there is a trade-off between mobility and the full functionality of intangible input interfaces. Output interfaces provide visual or non-visual feedback to the operator, e.g., through stationary displays or head-mounted displays that enable augmented reality environments [11]. In contrast to the most used user interfaces, which are primarily based on manual, tangible input and visual output, voice user interfaces (VUI) are a subset of HMI that enable voice-based

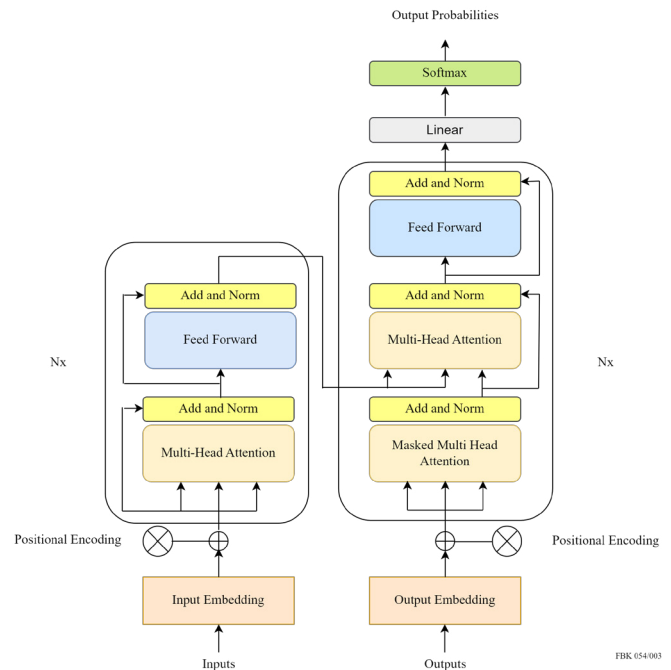


Fig. 1. The transformer architecture as developed in [6]. The Multi-Head Attention and the Positional Encoding of the words help in deriving near-human semantic information from sentences.

interaction (input) with a spoken or text-based language application (output) or further output channels.

VUIs consist of different software and hardware components that are needed for their human-machine interaction. On the hardware side, parts are needed for recording the voice (e.g., microphones), providing generated answers (sound-based or visual-based), and the computational resources needed to process those inputs. On the software side, automatic speech recognition (ASR) and text-to-speech (TTS) synthesis are mandatory. Speech recognition converts spoken words into text for further processing and information extraction [12]. Due to the current developments on NLP (see Section 2.1) improvements in ASR are possible.

In addition, for generating audio-based feedback, an instance for TTS is needed. Although text-to-speech synthesis has been possible for decades, only new approaches based on unsupervised learning can provide a precision that is comparable to a human level [13]. Moreover, for intelligent interactions between operators and machines, adequate dialog management and action selection are required. VUIs lead to intuitive human-machine interaction due to using language as a natural input channel. However, VUIs are currently mainly adopted and deployed as intelligent personal assistants for consumer electronics and the healthcare sector

2.3. VUI in Manufacturing

Studies have been made in the recent past regarding the use of VUIs for controlling industrial cells and robots. Studies like [15] document the design of voice-controlled CNC machines which use external voice recognition engines for implementing

the VUIs. The voice recognition engine used here can work on a limited command set of 80 commands, which on one hand guarantees the safety of the operation by staying to a limited set of confirmed voice commands, but on the other hand, limit the full potential of the VUI. Studies like [2] consider the challenges of implementing a VUI for industrial scenarios. The accuracy of the VUI here is marked important as a wrong interpretation of the command can lead to erroneous machine operations which compromises the safety of the operator as well as the machine. In order to mitigate the safety concerns, the study tries to implement a voice command description (VCD) sublanguage for use in industrial cells.

Models built using current deep learning architectures like Transformers as pointed out in 2.1 provide near-human accuracy for NLU. [16] suggests a similar approach to this paper introducing a concept for VUI-based control of a CNC machine. It suggests the limitation of using a VUI as a substitute for the conventional HMI used for controlling CNC machines. Specific scenarios for training the NLP module used in the voice recognition system for industry-specific scenarios are quite important as it involves the rightful participation of the machine to validate the training procedure. This can be facilitated by using a Software in Loop approach (SIL) for overcoming such tasks. The NLP used in the VUI is intensively trained using a virtual twin of the CNC machines that can accurately replicate the behavior of the machine. This helps in the seamless training of the Large Language Models (LLM) for machine-specific scenarios.

The current article proposes a conceptual approach for building up a VUI using the state-of-the-art transformer architecture for NLP and SIL for catering to the fast training requirements and also as an interactive user interface for Operator 4.0 in the context of CPMT.

3. Architecture

3.1. Requirements and Objectives

The primary objective of implementing a VUI for controlling machine tools is to increase flexibility by facilitating intangible interactions. This cannot be achieved without fulfilling certain hardware and software requirements. A computing system capable of doing high-performance computing in the range of giga floating point operations (GFLOPS) mainly leveraged with the help of a Graphics Processing Unit (GPU) is needed for training the NLU architecture in the VUI pipeline. A high-performance server is needed to host the software which can communicate over a low-latency, high-throughput network. Additionally, high-definition headsets and virtual reality (VR) glasses are needed for communicating with the VUI and DT. This can be achieved by using 5G wireless network for communication between the VUI, the digital twin (DT), and the physical machine. This standard can provide a bandwidth of up to 10 Gbps and low latencies upto 1ms.

Complementary to the hardware requirements, special software and programming skills should be employed to implement the VUI. Commercially available speech recognition and gen-

eration modules can be employed as the I/O block for the VUI. Pre-trained NLU architecture can be re-trained for machine-specific command sequences. Special operating systems are needed for the edge server and also dedicated communication protocols like OPC UA or MQTT for communication between the machine, the DT, and the VUI.

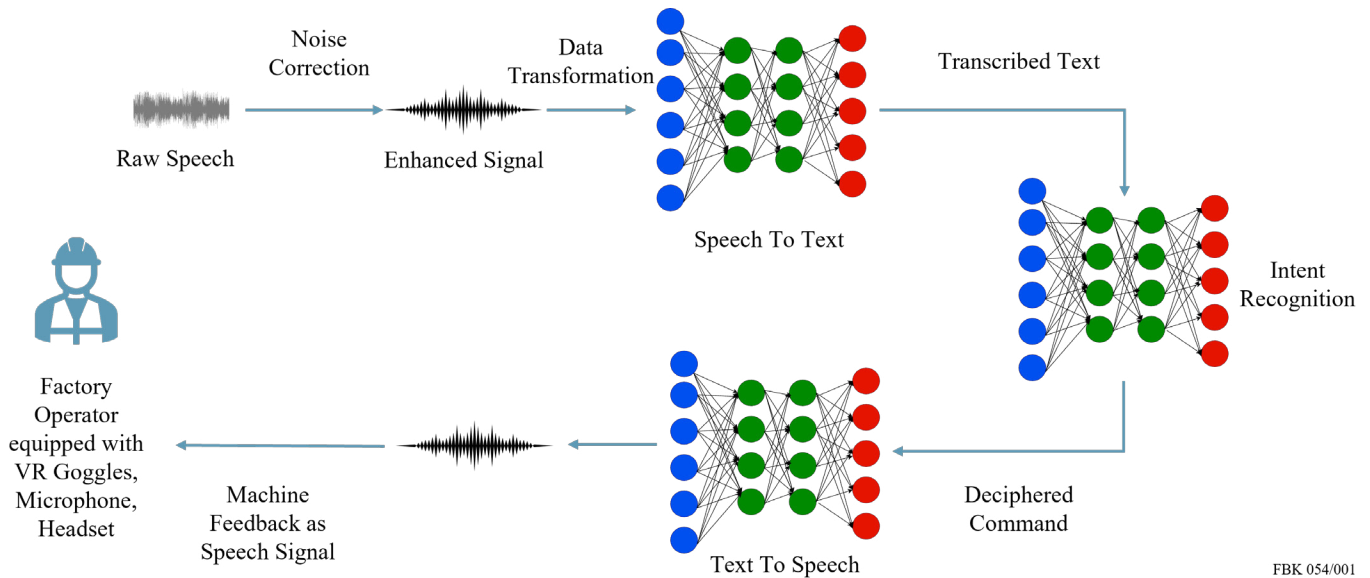
The applicability of the VUI in the industrial environment is dominated by safety demands. The safety of the operator and the machine needs to be keenly analyzed before such a solution is deployed in a real-time scenario. Emergency scenarios need to be evaluated with respect to the response time of the VUI once an emergency stop or hold command is initiated for the machine. Proper feedback should be formulated for a clear and concise understanding of the output of the machine by the operator. Special alternative routines need to be set in place to identify malintent on the part of the operator, for example, the VUI should contain logic for identifying dangerous operational sequences, that can cause damage to the tool or the shop floor and handle them with safety first guidelines.

Another very important requirement for the use of the VUI is the desensitization of the data wherever needed. The feedback from the VUI should be formulated in such a way that no personal information regarding the operator is read out or displayed. Similarly, special data protection guidelines should be followed whenever using sensitive information pertaining to the machine or the operator, for example, commands enabling diagnosis of the machine tools should not be available to the operator but only to the maintenance personnel. The VUI should assume role-based or responsibility-based commands from different users upon proper authentication.

3.2. Conceptual Pipeline

VUIs can be broadly broken down into a pipeline of different tasks like STT, NLU, and TTS. A software-in-loop (SIL) pipeline can be conceptualized for training and testing VUI for industrial machines. Virtual commissioning of the physical hardware eradicates its usage and the latency-based limitations posed by the communication medium. This can help in training the NLU architecture in a considerable amount of time. This section deals with explaining the conceptual pipeline of such a VUI as is depicted in Fig. 2. The above-mentioned pipeline is only a broad conceptual definition of the VUI to be refined and implemented in future works with the definition of a concrete system architecture followed by the development of use case-specific functional and system requirements.

Specifications and user stories can be created after interaction with various end-users. These documents can help in devising the possible natural language command scenarios and their mappings that can be used for retraining the NLU modules. The recording data should be created under the supervision of domain experts and should also consider domain-specific background noises to resemble a factory background. This is made possible by recording background noises at end-user sites. The use of commercially available models for speech recognition puts a limit on the amount of recording data needed for training the DL architecture to the machine tool-specific use case.



FBK 054/001

Fig. 2. The different levels of implementation of a Voice-User-Interface(VUI).

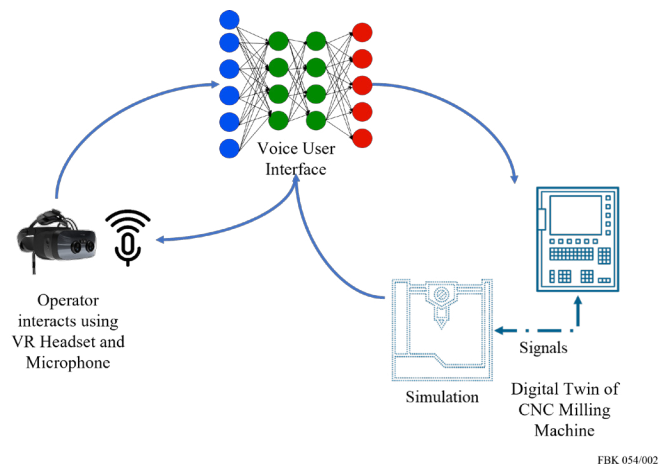
The ASR module converts the speech signal to transcribed texts. At this stage, the computer understands the speech only as language or dialect-specific characters without any semantic understanding of the text. Natural Language Understanding (NLU) is used to extract key phrases from the sequence of words transcribed by the ASR module. Advances in NLP have helped in achieving near-human performance in the semantic interpretation of texts. State-of-the-art NLP architectures called Transformers as introduced in Section 2.1 are used for semantic understanding or intent classification of the text. These architectures which have already undergone an unsupervised pre-training on large text-based datasets from the internet can be re-trained for machine-specific NLU tasks. Data collected from various information sources like machine tool-specific use cases, machine manuals, and domain knowledge of experts annotated with intention labels, is used for training the transformer architecture. The transformer architecture identifies the intent of the transcribed text and extracts the CNC command related key-phrases. These key phrases are mapped to command sets for the CNC module of the machine tool.

The confirmation and the feedback related to the machine operations are provided back to the user using conventionally available TTS architectures as per the accent and the language selected by the user. Usage of commercially available pre-trained ASR and TTS services provided by Amazon mitigates the difficulties faced with speech recognition and synthesis as they can be customized depending on accents, and speech rate and also provide near-human-level translations in the language choice of the user.

3.3. Implementation

This section gives an exemplary implementation of the above-defined conceptual pipeline on an application scenario of a CNC-enabled 3-axis milling machine. Fig. 4 gives a pictorial

description of the flow of data and processes in the application scenario. A sample neural network(NN) model diagram is used for representing the VUI here.



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Fig. 3. Software in Loop VUI Training and Testing of a 3 Axis CNC Milling Machine. The blue lines represent the flow of data through the pipeline. Digital twin adopted from [18].

"Hey CNC!" is the command for starting the dialogue with the machine. Such a word is also called a wake word and is often used in Voice Assistants to enable dialogues. A common example is "Hey Google" for Google Voice Assistant. The ASR module transcribes the wake word which is then extracted by the Transformer-based NLP architecture. On identifying the wake word, a certain feedback message is provided in the form of sound output by the TTS module of the VUI confirming the readiness of the software to accept the next command.

Upon issuing the wake word the operator can issue general commands like "Load .CAD file", "Start Milling", and "Check for workpiece". The text transcriptions of the voice

commands are then keyword extracted by the transformer-based intent classification module. These command parameters are then mapped to appropriate command phrases. An instance of such a mapping is the mapping of the text "Move the spindle left by 10 cm and down by 2 cm" to the key phrases "Move X Left 10 and Move Z Down 2". Apart from creating command mappings for correct command sets, mappings for incorrect command sets are also created and alternative strategies are employed for completion of such incomplete sets.

Specific feedback responses are developed that ask for confirmation of the command set depending on the inference of the command from the transcribed text. A confirmation message is sent back to the operator via the corresponding TTS interface before running the command. The operator can confirm the command and ask for a preemptive execution on the DT to analyze and find out possible failures leading to safety concerns. The operator uses VR glasses to view the operation of the virtual twin. Using such ways of interacting with the machine, the operator fulfills the role of the operator 4.0 working on a CPMT [1]. This execution can then be transferred to the real machine using specific voice commands.

Training an ASR, TTS, or NLP architecture from scratch is marked redundant due to advances already made in these areas. Data collected from sources like the machine manufacturers and end-users like the valid CNC command set is used for developing the DT and also training the NLU. The DT acts as the substitute for the physical machine during the process of training the NLU module in the VUI and also testing the performance of the VUI. The manufacturers and end-users help in providing domain-specific background noises that resemble a shop floor. Domain experts help the annotators in recording or devising commands specific to the milling machine. The experts also provide necessary information on valid and harmful command sequences, which can lead to the workpiece or tool damage. Data sensitive to machines are identified with the help of knowledge experts and anonymized to cater to the requirements of data protection.

3.4. Benefits and Challenges

One of the primary benefits of such an interface is the increased flexibility of human-machine interaction. VUI enables freedom of hands during operation. So machine tools can be interrogated and controlled by people who need their hands for other work processes (e.g. assembly, tool preparation, etc.). Moreover, a VUI can be used with dirty hands, with gloves, especially when the operator has to be careful about sterility. Machine tools can also be operated by people with disabilities who can not use their hands like other people, which leads to increased inclusion in manufacturing.

The safety of the operator is increased as the machine tool can now be operated from a safe distance and rather than a fixed console next to the machine. For example, a VUI can be used at a medium distance up to 5 meters and the operator does not have to step up to the operator terminal for a short interaction. Particularly in the case of multi-step manual processes, such as the changeover of a larger machine, great potential for effi-

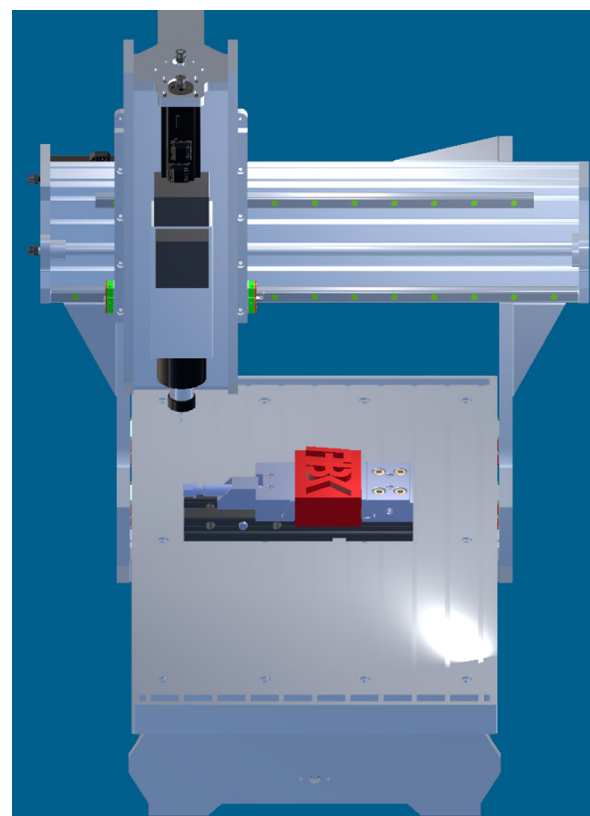


Fig. 4. Virtual Simulation of the 3 Axis CNC machine working used as the digital twin.

ciency is created if the interaction does not have to take place at a distant terminal between each step.

Additionally, voice control makes hardly any demands on space, cleanliness, or special lighting conditions and can therefore be used without restriction even in adverse working environments. Moreover, the flexibility and speed of the interaction can be increased by the combination of different manual command sequences to one speech-based command. Information exchange with a precise question or instruction can be given more quickly than manually navigating dialog menus and using different keystrokes.

Next to the benefits, there also exist challenges in implementing the proposed concept. The accuracy of the STT, TTS, and NLU modules poses the greatest challenge in implementing the VUI in an industrial scenario. It should be made sure with running specific and intensive test cases that the prediction modules work as close to the human level as possible and that errors if propagated are properly handled at respective levels in the software architecture. An inappropriate mapping of the intended and executed command can lead to erroneous operation which can be harmful for the tool as well as the operator.

The real-time suitability of the VUI also poses a major challenge in its acceptance in industrial scenarios. The response time of the VUI can seem vital in emergency scenarios where an immediate stop or halt of the machining is needed for example in case of tool damage or fire. Modules catering to emer-

agency routines should be carefully designed to minimize response times and alternate mechanisms like manual override should be kept in place to ensure higher safety. Low latency of the VUI can also come as a challenge in the industrial scenario as usage of high bandwidth networks like 5G, or a high-performance edge server for hosting the software can be expensive. The independent modules of the VUI should be implemented keeping such requirements in mind, for example, a trade-off should be considered between the model size of the LLM to be used and the hardware storage and computation requirements of the VUI.

4. Conclusion and Outlook

The proposed concept enables the implementation of VUI for manufacturing, especially for the control of machine tools. Due to the utilization of NLPs that are trained with data sets that contains specific technical information and expressions, intuitive human-machine interaction will be realized. For the training of the algorithm, a virtual model of the machine tool is utilized in an virtual reality environment. By integrating the specific hardware for the VUI (e.g. microphone, speaker, etc) for a hardware-in-the-loop development. This leads to flexible and fast training of the NLP for specific, machine tool related tasks.

The implemented VUI leads to intangible, flexible, inclusive, and intuitive human-machine interactions that can cope with the high requirements of modern CPPS. However, user acceptance, integration of implicit information of operators, and the robustness of the overall system have to be investigated. For future research, the VUI will be implemented and evaluated in a real manufacturing system with the machine tool described in Section 3.3. Moreover, the VUI will be adapted and implemented for interaction with other machine tools - even simultaneously - as well as for the control of further processes (e.g. automated guided vehicles, robotics, warehouses) in manufacturing systems.

Acknowledgements

This research is funded by the German Federal Ministry for Education and Research (Bundesministerium für Bildung und Forschung, BMBF, "Gefördert vom BMBF") within the project "KMU-Innovativ - Verbundprojekt DialoKIM: Dialogorientierte Künstliche Intelligenz zur intuitive Maschinesteuerung" (Grant number 01IS23008C).

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